Cycling Safety Reimagined: A Transperant Approach to Pedestrian Alerts

ISE 5654—Human Factors in System Design Project

Catalina Reinhart
Industrial and Systems Engineering
Virginia Tech
Blacksburg, VA
catalinar@vt.edu

Astha Dhakal
Computer Science
Virginia Tech
Blacksburg, VA
asthad@vt.edu

Asa Castleberry
Industrial and Systems Engineering
Virginia Tech
Blacksburg, VA
aceberry73@vt.edu

Andrew Stewart
Industrial and Systems Engineering
Virginia Tech
Blacksburg, VA
andrewcstewart@vt.edu

I. Introduction

In today's vibrant urban environments, the shared use of pathways by cyclists and pedestrians has introduced significant safety challenges. With the rise of advanced audio devices, features such as Transparency Mode in modern headphones aim to allow ambient sounds to pass through while enabling users to enjoy their music. However, its effectiveness in alerting pedestrians to immediate dangers—like approaching cyclists—remains uncertain. Traditional bicycle bells, once effective in alerting pedestrians, now struggle to penetrate the sound barriers created by these advanced devices, thereby heightening the risk of accidents. Our project seeks to address this critical issue by developing an innovative warning system for cyclists, utilizing state-of-the-art Bluetooth speaker technology. The critical questions that will lead our experiments would be:

- How well do Bluetooth speaker sound modulation techniques work to get around headphones with transparency mode?
- What specific adjustments in pitch, roughness, irregularity, and power result in the greatest improvements for alerting pedestrians?
- How do reaction times and accuracy vary among pedestrians when exposed to different sound profiles?
- What roles do personal preferences, and sound associations play in the effectiveness of these alerts?

By investigating these questions, we aim to design a system that can reliably alert pedestrians, even those wearing modern earbuds. Our methodology will include thorough evaluations of reaction times, accuracy, personal preferences, and sound associations to optimize the system's performance [10]. This solution not only seeks to minimize the risk of collisions but also has the potential to transform the auditory experience in shared urban spaces, creating a safer environment for both cyclists and pedestrians. Through this research, we aspire to establish a new benchmark in urban mobility safety, ensuring that advancements in personal audio technology do not compromise public safety.

II. Related Work

Traditionally, bicyclists use bells to warn pedestrians of their presence. With advancements in Bluetooth sound speaker technology, modern bells can have speakers and music playing capabilities. However, the popular trend of active noise-canceling (ANC) headphones can reduce the effectiveness of these acoustic warning devices. The effectiveness of bicycle warning devices communicating with earbud wearing pedestrians is currently undeveloped [1,2]. Additionally, the popularization of E-bike means some bicyclists are moving at faster speeds and require larger safety distances [3][4].

Bicyclists often have trouble signaling their presence to nearby pedestrians, especially in crowded areas where pedestrians are wearing headphones [5]. As a result, pedestrians are injured and killed [6]. One study found that near accidents occur 50 times more frequently than collisions, meaning the use of headphones could turn some of the near accidents into collisions [7]. Bicycle and E-bike use is also promoted as a sustainable mode of transportation. Therefore, the use of headphones while traveling and the promotion of faster traveling bicycle use indicates more pedestrian-bicyclist interaction research needs to be conducted.

Most bicyclist bells use traditional bell dings and horns which have mixed results without the addition of ANC headphones into bicyclist-pedestrian interactions [2]. Instead, Frohmann et al. found that urgency and recognizability are the two most important and intertwined parameters in bike bell perception [2]. One key example is the digital bike bell in that study found participants did not perceive the digital sound to be indicating an approaching bicyclist. An HFES article has investigated a solution by making a smart digital bike bell that provides localization information and polite/rude sounds [8]. However, there is no established solution to bike bell urgency, recognizability, and localizability in literature. Additionally, no published studies we found sought to create a bicycle bell capable of overcoming earbud-wearing pedestrians.

This study will fill a knowledge gap in bicycle bell technology to address modern technology and pedestrian habits. This study will also provide theoretical and practical contributions. Our study will add to the existing bike bell sound models by accounting for bicyclist communication with headphone wearers in transparency mode to increase shared path safety. As such, this study will practically contribute to pedestrian and bicyclist safety. Future research can also implement the sound profiles created in this study into products.

III. Task Analysis

A. Goals

Our system is designed to alert pedestrians to the presence of cyclists in busy areas where people use headphones in transparency mode frequently. To effectively overcome the challenges posed by these headphones, our alerts will be engineered to ensure that sound frequencies can penetrate the noise cancellation. This will involve variations in pitch, roughness, intensity, and irregularity. Additionally, our interface will enable cyclists to indicate the direction from which they are approaching, further enhancing pedestrian safety.

B. Tasks

The key task we will focus on is the activation of the warning signal, which constitutes the most common function of our system. This task is vital, as it involves notifying pedestrians of an approaching cyclist to help prevent potential collisions. We chose to prioritize this scenario because it represents the primary use case for the device. The process begins when a cyclist approaches an area and encounters a pedestrian who may be using active noise-canceling headphones. The cyclist then needs to determine the best way to alert the pedestrian while ensuring a safe passage. Our device is designed to generate a signal that can penetrate these noise-canceling barriers, ultimately promoting safety for both the cyclist and the pedestrian. The task analysis for the project is depicted in the diagram.

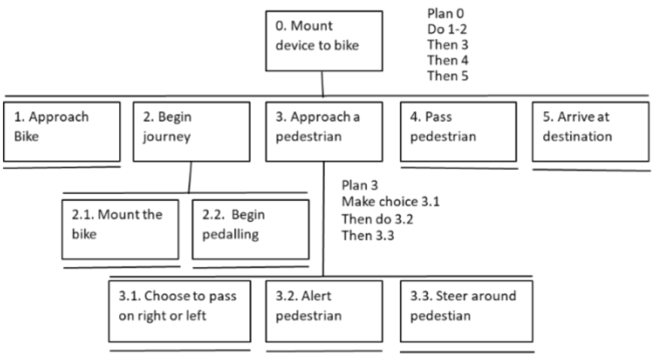

Figure 1: Task Analysis

C. Design Implications

Table 1: Design Considerations and Implications

Task Analysis Element/Plan	**Design Considerations/Implications**
1. Users may be wearing headphones in transparent mode; thus, unable to hear warning signal (Task 3.2)	Create a sound that has a frequency that surpasses the transparent mode features.
2. Signaling “on your right” or “on your left” confusion (Task 3.1)	Sounds will need to be distinct for pedestrians to understand the difference between the two signals. The position, shape, and color between the two different buttons will need to be distinct for users to differentiate between the two.
3. Reach of the signal button (Task 3.1)	Placement of the bell must be strategic so that users will not have to extensively reach to hit the signal.

IV. Methods

A. Experimental Design

This experiment aimed to explore the communication gap in current bicycle bell technology to properly alert the modern pedestrian utilizing noise-canceling headphones. Given the timeframe, a within-subject design with 12 participants was the best course of action for this experiment. This helped to eliminate between-subject variability, allowing researchers to observe the changes in behavior over time with a smaller sample size. Participants were exposed to two primary conditions: walking without earbuds and walking while wearing earbuds playing a pre-selected song, “What Once Was” by Her’s, at 60 dB (approximately 30% volume on AirPods). To further simulate real-world scenarios, an additional variable—Transparency Mode—was introduced to mimic headphone features that allow ambient noise to pass through. Under these conditions, participants encountered four auditory signals: two traditional bell-like sounds and two novel sounds specifically developed by the researchers. The auditory cues were standardized in loudness and designed to play instantly upon activation to ensure consistent timing throughout the trials.

When representing the bicyclist warning, participants engaged with four different sounds: two traditional bell-like sounds and two novel sounds created by the researchers. To measure the effectiveness of these sounds in their ability to surpass noise-canceling technologies, subjective and objective measures were collected and analyzed. A videotaping was conducted to properly observe the behaviors of each participant before, during, and after each sound. These behaviors included the movement of the participant (i.e. moving to right, left, if they were startled, if they did not move, etc.) as well as the reaction time from the onset of the auditory cue to the point where the participant made a move. After each sound, an audio user experience questionnaire was provided to further analyze the impact that each sound had on the participant.

This questionnaire can be found in Appendix A, Additionally, a satisfaction survey was given to each participant at the end of the experiment to measure the overall user experience. The questions for the satisfaction survey are laid out in Appendix A.1.

B. Stimuli Design and Equipment

The user interface for bike sounds was auditory in nature. A computer was used as a “Wizard of Oz” control interface for the bike sound device. The design innovation came from the novel sounds we made that notified a pedestrian that a bicyclist was passing them.
Figure 1 depicts our experimental set-up. The person on the left was the participant, and for this demonstrated trial, was using headphones in transparent mode as depicted by the small crescent symbol. The person on the right was a researcher. Both the participant and researcher were on a walking path. The walking path was created with boundaries defined using blue masking tape. The boundaries were made to classify participants' movement previously described. The researcher used a laptop to play one of our stimuli conditions as depicted in Table 1. The stimuli conditions were created in Audacity software. The researcher on the side recorded the participant’s reaction using a phone camera. Recording the experimental trials allowed for an objective post hoc analysis of participant reaction movement and reaction time. Finally, the desk and additional laptop had a survey on it that the participant completed after trials as shown in the bottom left of figure 1.

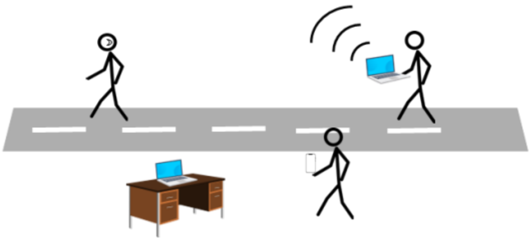

Figure 1: User Interface

Table 1: Stimuli Conditions

	Sound 1	Sound 2	Sound 3	Sound 4
Stimuli				
Sound Description	Novel sound #1 has different roughness and other audio properties. Has a “da-da” sound.	Sound resembles a traditional bike bell. Has a “ding” sound.	Novel sound #2 has a different roughness and other audio properties than the other sounds. Has a “chime” sound.	Sound resembles a traditional bike horn. Has a “ring- ring” sound.

The procedure used for experiment are described below:

1. Pre-Experiment Setup: The environment was set up to simulate a crosswalk using paper or chalk on the floor. A person holding a computer acted as the "biker" in the simulation. Audio playback equipment was prepared to play the 4 sounds at randomized intervals.

2. Participant briefing: Participants were greeted and informed that they were part of a study on pedestrian responses to bike sounds. They were instructed to react naturally to any sounds they heard as they walked across the simulated road. It was explained that they would experience four different sounds and that they would complete a brief survey for each sound after each set of four trials.

3. Walking Trials: Participants walked along the simulated road while the "biker" played one of the sounds. Participants responded to the sound naturally (e.g., by stopping, looking, moving left or right). The reaction time from the onset of the sound to their movement was recorded using a stopwatch. Video recordings captured their behavior for later analysis.

4. Headphone conditions: Participants experienced each sound both with headphones (transparent mode) and without headphones, with the order counterbalanced across participants.

5. Breaks: No formal breaks were scheduled in the experiment. However, participants were free to take breaks as needed.

6. Surveys: After each round of four sounds, the participants took the BUZZ Audio UX scale questionnaire. See **Appendix A**. to see the questionnaire.

7. Post-Experiment Data Collection: Video footage was analyzed to measure the reaction time in milliseconds. The direction of the pedestrian movement (left, right, or no movement) was documented. Observable startle responses or hesitations were also noted.

8. End of session: Participants were thanked for their involvement and debriefed on the study's purpose. The participants took an open-ended satisfaction questionnaire to capture their general thoughts. All data was saved, and the equipment was sanitized before the next participant would use it.

B. Participants

The study involved 12 participants (6 male and 6 female) ranging in age from 21 to 57 years. Participants were recruited to represent a diverse sample of pedestrians in realistic urban walking scenarios. This gender balance and age range were chosen to capture a variety of natural walking behaviors and responses to auditory stimuli.

V. RESULTS

The focus of this study was to evaluate the differences between “transparency mode” headphones and bike bell sounds affected response movement and reaction time. The Buzz sound questionnaire was provided to participants after every trial was analyzed to investigate subjective ratings of the different bike bells. The rejection criteria for the purposes of this data analysis were set to p = 0.05.

Both response movement and reaction time had a nonnormal distribution according to Shapiro-Wilk goodness of fit tests. However, ANOVA is robust to nonnormal conditions. Therefore, a two-way ANOVA was performed to evaluate response movement and reaction time between sound and headphone condition. Response movement did not differ between sounds or headphone condition as seen in Table 2. However, reaction time did differ between the sound conditions (Table 2). Sound 1 ($p = 0.0092$) and Sound 2 ($p = 0.0087$) significantly increased reaction time. Sound 3 ($p < 0.0001$) significantly decreased reaction time.

Table 2: Mean (SD) values of response movement and reaction time by sound and headphone condition.

	Headphone condition	Response Movement	Reaction Time	Mean Reaction Time
Sound 1	On	2.25 (0.77)	0.80 (0.27)[t]	0.86 (0.43)
	Off	2.06 (0.79)	0.92 (0.54)[t]	
Sound 2	On	2.17 (0.70)	0.84 (0.24)[t]	0.86 (0.33)
	Off	2.25 (0.73)	0.89 (0.41)[t]	
Sound 3	On	2.19 (0.89)	0.77 (0.39)	0.78 (0.40)
	Off	2.31 (0.82)	0.79 (0.41)	
Sound 4	On	2.47 (0.74)	0.56 (0.21)[t]	0.57 (0.22)*
	Off	2.31 (0.86)	0.59 (0.22)[t]	

*Sound condition is statistically significant from other sound conditions(p=0.05)
[t]Sound condition is statistically different from overall reaction time mean (p=0.05)

Sound 4 had a significantly different reaction time than the three sounds via post hoc t-tests (Tukey HSD test) as shown in Figure 2. There was no interaction between sound or headphone condition.

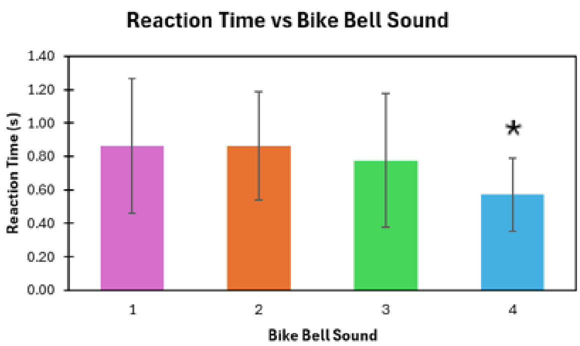

Figure 2: Mean reaction time between sound condition. Sound four is statistically significant (*) via Tukey HSD test (post hoc t-tests). Sound 2 is different from sound 4 ($p<0.0001$), Sound 1 is different from Sound 4 ($p<0.0001$), and Sound 3 is different from Sound 4 ($p=0.003$). Each error bar is constructed using 1 standard deviation from the mean.

While the Buzz Audio Questionnaire Likert-scale data is ordinal, the parametric tests Analysis of Variance and post hoc t-tests were performed. This was chosen as previous work has suggested that t-tests provide similar data for nonparametric Likert-scale data as nonparametric tests [12, 13]. Analysis of Variance revealed sound was a significant predictor for all questions ($p<0.05$) except for the pleasant, fun, and boring questions.

Post hoc analysis was performed (as shown in Figure 3) and found Sound 4 significantly better perceived scores in Helpful, Understandable, Reliable, Confusion, Meaningful, and Distinguishable questions by at least 35% when compared to the other three sounds. Sound 4 had a better score for Difficult than Sound 1 and 2. Sound 2 significantly decreased perceived interest compared to the other sounds, and Sound 2 had higher perceived meaning than Sound 1. The headphone condition was not a significant predictor. Additionally, there was no interaction between sound and headphone condition for any of the responses. The descriptive statistics are available in Appendix A.3.

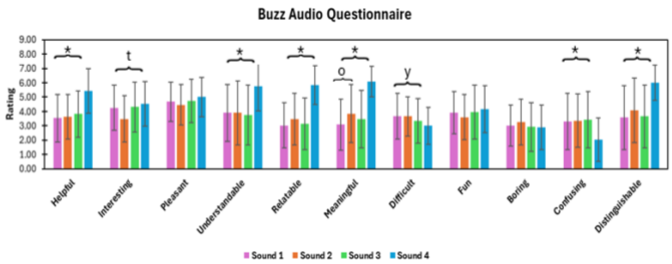

Figure 3: Buzz Audio Questionnaire Mean responses for each bike bell sound. Difficult, Boring, and Confusing are negatively worded statements, where a lower rating is better. *Significant Sound 4 rating difference from the other three sounds. [y]Significant Sound 4 rating difference from Sounds 1 and 2. [t]Significant Sound 2 rating difference from the other three sounds. [0]Significant Sound 2 rating difference from Sound 1.

Analysis of Variance found the ending survey question, "Please rate your overall experience with the sounds on a 1-5 scale. (1: worst, 5: best)," to differ by bike bell sound. Post hoc t-test analysis revealed Sound 4 had the best overall experience ($p<0.0001$) compared to the other three sounds. Figure 4 has a further t-test sound differentiation. The descriptive statistics are available in Appendix A.4.

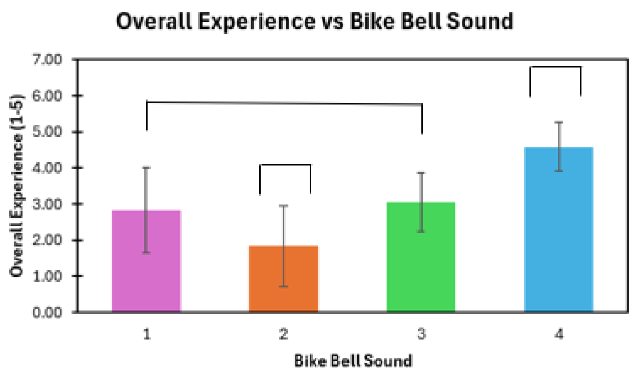

Figure 4: Mean Overall experience rating 1-5 with 1 being the worst and 5 being the best. Any sounds not connected by a common bracket are significantly different from each other ($p<0.05$).

VI. Discussion

In this study, we found that reaction times were dependent upon whether the user was experiencing a transperant headphone system or not. Those that were wearing the system produced lower reaction time values than those without. Users preferred sounds that were closely related to traditional bike sounds, even if their reaction time values were the lowest with their top sound preference. Many users emphasized their interest in familiar sounds, leaning towards more traditional bike sounds. These results are important because they provide insight on the user safety experience of pedestrians wearing audio devices with transparent settings. Overall, our study contributes to the understanding of auditory interfaces within urban environments where a user will have to interact with safety alert systems such as those associated with cyclists.

Our findings build upon previous studies. With our satisfaction surveys for each sound, we confirmed the findings of Frohmann et al. on how urgency and recognizability are the most imperative characteristics of bike bell perception [2]. With **Sound 3**, 8 out of our 12 participants thought the sound was inappropriate for the situation. This is due to their perception of the sound being electronic, as if indicating a transaction or a notification from a phone. With all the possible audio cues from a user's phone—which is typically the device linked up to their headphones—participants stipulated that their main confusion is caused by the thought of the bike bell being a notification from their phone.

The findings of the study also align with the current theory behind auditory alarm recognition. Social Constructivism is a theory that states that our understanding and interpretation of the society we are a part of is shaped by taught social and cultural norms. This theory is seen in how different cultures use different alarms for national and local alerts, as well as emergency personnel, that are understood by that specific culture yet are new to foreign populations and are not understood as quickly. The phenomenon may also be explained by schema theory, which states that people develop mental schema based on their prior experiences. This, in short, means that people interpret the sounds that they previously associated with bike bells to be a warning of an oncoming cyclist.

This knowledge can be utilized to further advancements in auditory interfaces and cutes. This can be specifically implemented within the realm of transparent headphones and safety environments. Our findings suggest that transparency mode needs further development before it can be advertised as a safe option for headphone users.

The research conducted will help to facilitate how we should move forward with research into the development of an electronic bike bell. Some of the considerations include if a novel sound can be developed that replicates a traditional bike bell sound. If this can be developed successfully, it should be more recognizable to the general population. Another consideration is if the bike bell itself should be designed to pierce through musical occlusion or if the headphones themselves should be designed to allow certain alarm sounds through for the safety of the user. One solution that could mitigate the need to design a piercing bike bell at all would be through bone conduction, which is an area of technology that is already seeing significant growth and development. The research conducted is important to the field of engineering though allowing our team to test different conditions of interference with one's auditory perception and to validate current theory regarding alarms and auditory interfaces.

VII. Conclusion

This study explored the role of auditory alerts in overcoming the challenges posed by noise-canceling headphones, a critical issue in ensuring safe cyclist-pedestrian interactions within urban environments. By evaluating both traditional bike bell sounds and novel, optimized auditory cues, we observed significant differences in reaction times and user perceptions. Specifically, **Sound 4** emerged as the most effective alert, demonstrating a marked reduction in reaction times while receiving high ratings for clarity, reliability, and overall user satisfaction. These results highlight the potential for sound design to address modern technological barriers, such as active noise-canceling headphones, which diminish the effectiveness of conventional auditory alerts.

From a human-computer interaction (HCI) perspective, this research advances our understanding of auditory feedback systems and their role in real-world safety applications. The design of **Sound 4** emphasizes principles of urgency, recognizability, and user-centered design, ensuring compatibility with diverse pedestrian behaviors and preferences. These insights demonstrate the importance of tailoring auditory signals to context-specific challenges, bridging the gap between emerging personal audio technologies and public safety systems.By leveraging novel auditory solutions, this study lays the groundwork for future research in multi-sensory interaction and adaptive alert systems. The findings contribute to the broader goal of designing safer, more inclusive urban spaces through the thoughtful integration of auditory technologies, reinforcing the potential of HCI to address pressing societal challenges in mobility and public safety.

A. Limitations

The study has several important limitations that warrant acknowledgment. First, the small sample size (n=12) limits the generalizability of our results. A larger and more diverse participant group is necessary to capture a broader range of user behaviors, preferences, and perceptual differences, offering a more comprehensive understanding of pedestrian responses to auditory cues. Another key limitation lies in the controlled lab setting. While the simulated environment provided valuable initial insights, it fails to replicate the complexity of real-world urban spaces. Factors such as unpredictable background noise, multiple distractions, and dynamic interactions between pedestrians and cyclists could not be fully captured in our current setup.

Additionally, the study's hardware selection was constrained to Apple AirPods tested at a single volume level. This narrow scope limits our findings, as variations in headphone brands, noise-cancellation technologies, and listening conditions may produce different outcomes. The auditory cue set, though informative, was intentionally limited to four types. This does not encompass the full range of potential sound variations that could influence pedestrian reactions and user satisfaction. Further exploration of diverse acoustic parameters is needed to optimize auditory alert effectiveness. Lastly, the method of manually tracking reaction times using video analysis introduces potential measurement inaccuracies. Future research should employ automated tracking tools or advanced behavioral measurement systems to enhance precision and reliability.These limitations, however, serve as opportunities for growth. They highlight clear directions for future research to expand and validate our findings, ultimately paving the way for more robust and effective auditory alert systems that address the complexities of real-world urban environments.

B. *Future Work*

To build on the insights gained from this study, future research should focus on enhancing bicycle warning systems through improved sound design, adaptive technologies, and user-centered innovation. Further development of sound profiles could involve creating advanced modulation techniques specifically aimed at penetrating active noise-canceling headphones. This would include developing a robust library of optimized sound frequencies capable of effectively alerting pedestrians across a variety of urban environments and user groups.

A key area for future exploration involves integrating technology to prototype dynamic warning systems that can adapt to different headphone technologies in real time. By incorporating smart city infrastructure, these systems could deliver intelligent, context-aware alerts, improving responsiveness and safety for shared pathways. From a research design perspective, future studies should include a larger, more diverse participant pool to account for differences in hearing ability, age, and auditory perception. Additionally, testing under real-world conditions—such as unpredictable ambient noise and varying pedestrian densities—will yield more realistic and practical insights into the effectiveness of these auditory alerts.

Another promising direction is the creation of adaptive auditory systems that can adjust based on variations in headphone transparency modes and levels of noise cancellation. By leveraging machine learning algorithms, these systems could dynamically analyze acoustic challenges and adjust alert signals in real time to maximize their effectiveness.

Addressing these research directions will contribute to a deeper understanding of how pedestrians respond to auditory cues in complex environments. Ultimately, these advancements will support the development of intelligent, scalable, and human-centered safety solutions that promote safer interactions between cyclists and pedestrians in modern urban mobility systems.

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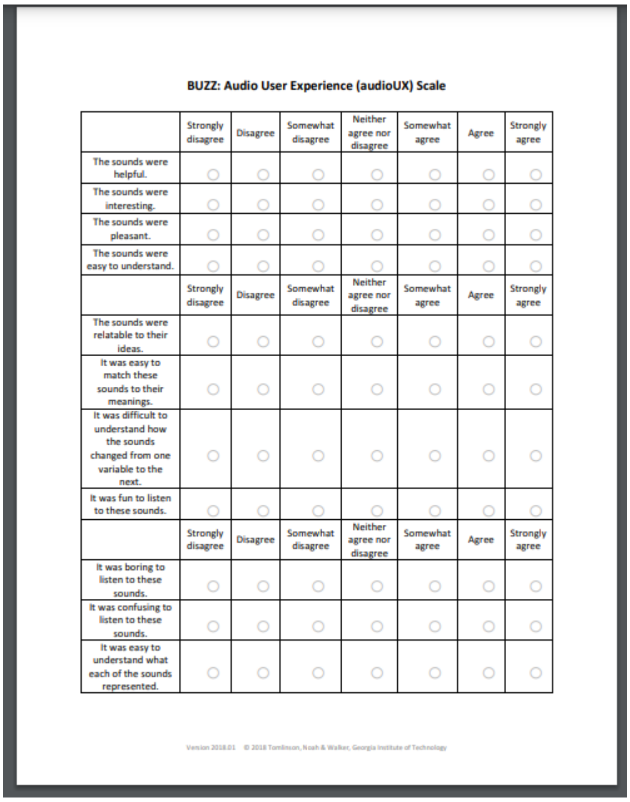
APPENDIX A

A.1 Screenshot of the BUZZ Audio UX scale questionnaire:



BUZZ: Audio User Experience (audioUX) Scale

	Strongly disagree	Disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Agree	Strongly agree
The sounds were helpful.	○	○	○	○	○	○	○
The sounds were interesting.	○	○	○	○	○	○	○
The sounds were pleasant.	○	○	○	○	○	○	○
The sounds were easy to understand.	○	○	○	○	○	○	○
	Strongly disagree	Disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Agree	Strongly agree
The sounds were relatable to their ideas.	○	○	○	○	○	○	○
It was easy to match these sounds to their meanings.	○	○	○	○	○	○	○
It was difficult to understand how the sounds changed from one variable to the next.	○	○	○	○	○	○	○
It was fun to listen to these sounds.	○	○	○	○	○	○	○
	Strongly disagree	Disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Agree	Strongly agree
It was boring to listen to these sounds.	○	○	○	○	○	○	○
It was confusing to listen to these sounds.	○	○	○	○	○	○	○
It was easy to understand what each of the sounds represented.	○	○	○	○	○	○	○

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A.2 Satisfaction Questionnaire Questions:

1. How did the sounds make you feel?
2. Do you feel that your reaction was appropriate?
3. Please rate your overall experience with the sounds on a 1-5 scale. 1 being worst and 5 being best.

A.3 Table of BUZZ Audio UX Questionnaire Data (mean [SD])

Buzz Survey														
Condition	**Sound**	**Reaction time**	**Response Movement**	**The sounds were helpful.**	**The sounds were interesting.**	**The sounds were pleasant.**	**The sounds were easy to understand.**	**The sounds were relatable to their ideas.**	**it was easy to match these sounds to their meanings**	**It was difficult to understand how the sounds changed from one variable to the next.**	**It was fun to listen to these sounds**	**It was boring to listen to these sounds.**	**It was confusing to listen to these sounds.**	**It was easy to understand what each of the sounds represented.**
on	1	0.805 (0.266)	2.25 (0.77)	3.583 (1.811)	4.083 (1.442)	4.694 (1.305)	3.917 (2.17)	2.972 (1.576)	3.194 (1.849)	3.694 (1.636)	3.806 (1.451)	2.917 (1.402)	2.889 (1.924)	3.389 (2.271)
	2	0.838 (0.242)	2.167 (0.697)	3.333 (1.474)	3.389 (1.609)	4.444 (1.319)	3.583 (2.347)	3.222 (1.869)	3.556 (2.104)	3.667 (1.414)	3.472 (1.612)	3.028 (1.63)	3.167 (1.765)	3.944 (2.292)
	3	0.769 (0.39)	2.194 (0.889)	3.833 (1.577)	4.278 (1.614)	4.611 (1.379)	3.611 (2.032)	3.028 (1.647)	3.528 (2.007)	3.5 (1.483)	3.75 (1.779)	2.944 (1.866)	3.611 (1.961)	3.472 (2.249)
	4	0.555 (0.214)	2.472 (0.736)	5.667 (1.352)	4.472 (1.699)	5.056 (1.433)	5.722 (1.632)	5.694 (1.508)	6.028 (1.108)	3.056 (1.241)	3.889 (1.489)	2.75 (1.519)	2.056 (1.433)	5.833 (1.32)
off	1	0.923 (0.545)	2.056 (0.791)	3.5 (1.502)	4.417 (1.713)	4.667 (1.414)	3.889 (1.785)	3.083 (1.556)	2.972 (1.715)	3.639 (1.57)	4 (1.493)	3.111 (1.469)	3.694 (2.012)	3.75 (2.222)
	2	0.891 (0.405)	2.25 (0.732)	3.944 (1.62)	3.583 (1.61)	4.472 (1.483)	4.222 (2.085)	3.722 (1.717)	4.139 (1.93)	3.639 (1.355)	3.722 (1.523)	3.472 (1.558)	3.556 (1.949)	4.222 (2.205)
	3	0.785 (0.412)	2.306 (0.822)	3.806 (1.653)	4.361 (1.869)	4.833 (1.63)	3.889 (2.162)	3.25 (1.948)	3.389 (1.975)	3.222 (1.623)	4.139 (1.973)	2.889 (1.489)	3.222 (1.944)	3.833 (2.131)
	4	0.587 (0.22)	2.306 (0.856)	5.194 (1.721)	4.611 (1.42)	4.944 (1.286)	5.75 (1.73)	5.972 (1.158)	6.139 (1.046)	2.944 (1.351)	4.444 (1.796)	3.056 (1.567)	2.028 (1.576)	6.167 (1.159)

A.4 Table Containing End Survey Data

Sound	Please rate your overall experience with the sounds on a 1-5 scale. 1 being worst and 5 being best.											Mean	S.D.
1	2	2	1	3.5	2	3.5	2	2	3	4	5	4	2.83	1.17
2	1	1	1	1	4	2	1	2	1	2	4	2	1.83	1.11
3	2	3	4	4	2	4	3	2	2.5	3	4	3	3.04	0.81
4	4	5	5	4	5	4	5	5	5	3	5	5	4.58	0.67